SIMATS Online Programming Lab

PYTHON PROGRAMMING

PATAN MOHAMMED RIYAZ KHAN
192572126RC

Q1 **Find smallest and largest number** 10 marks

WAP to find smallest and largest number in an array

input: 12 5 18 3 25 9

output:

Smallest number:3

Largest number:25

Your Code

```
arr = list(map(int, input().split()))

smallest = arr[0]
largest = arr[0]

for i in arr:
    if i < smallest:
        smallest = i
    if i > largest:
        largest = i

print("Smallest number:", smallest)
print("Largest number:", largest)
```

Input

```
12 5 18 3 25 9
```

Output

```
Smallest number: 3
Largest number: 25
```


Run Save Submit

Q2 **Missing Number** 10 marks

WAP to find missing number in an array of shuffled order

input: 4 2 1 6 5

output:3

Your Code

```
arr=list(map(int,input().split()))
n=len(arr)+1
for i in range (1,n+1):
    if i not in arr:
        print(i)
        break
```

Input

```
4 2 1 6 5
```

Output

```
3
```

Run Save Submit

Q3 **Second largest number** 10 marks

WAP to find second largest number in an array

INPUT: 12 5 18 3 25 9

OUTPUT: 18

Your Code

```
arr = list(map(int, input().split()))
arr.sort()
print(arr[-2])
```

Input

```
12 5 18 3 25 9
```

Output

```
18
```

Run Save Submit

Q4 **Most repeated number** 10 marks

WAP to print the most repeated number in an array

INPUT: 2 4 6 2 5 3 1 2

OUTPUT:2

Your Code

```
arr = list(map(int, input().split()))

freq = {}
for x in arr:
    freq[x] = freq.get(x, 0) + 1

most_repeated = max(freq, key=freq.get)
print(most_repeated)
```

Input

```
2 4 6 2 5 3 1 2
```

Output

```
2
```

Run Save Submit

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Q5 **Unique elements** 10 marks

WAP to print all unique elements in an array

INPUT : 2 4 6 2 5 6 7

OUTPUT : 4 5 7

Your Code

```
arr = list(map(int, input().split()))
freq = {}

for x in arr:
    freq[x] = freq.get(x, 0) + 1

for x in arr:
    if freq[x] == 1:
        print(x, end=" ")
        freq[x] = 0
```

Input

```
2 4 6 2 5 6 7
```

Output

```
4 5 7
```

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Q6 **Rotate an array by 'N' positions** 10 marks

WAP to rotate an array by 'N' positions

INPUT: 2 5 6 1 3 4

OUTPUT: 6 1 3 4 2 5

Your Code

```
def rotate_array(arr, n):
    n %= len(arr)
    return arr[-n:] + arr[:-n]

data = list(map(int, input().split()))
n = data[0]
arr = data

print(*rotate_array(arr, n))
```

Input

```
2 5 6 1 3 4
```

Output

```
3 4 2 5 6 1
```

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Q7 **Remove all duplicates in an array**

10 marks

WAP to remove all duplicates in an array

INPUT: 2 4 6 2 7 8 3 4

OUTPUT: 2 4 6 7 8 3

Your Code

```
arr = list(map(int, input().split()))

seen = set()
result = []

for x in arr:
    if x not in seen:
        seen.add(x)
        result.append(x)

print(*result)
```

Input

```
2 4 6 2 7 8 3 4
```

Output

```
2 4 6 7 8 3
```

Run Save Submit

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Q8 **Print all 0's followed by non-zero without changing the order** 10 marks

WAP to print all 0's followed by non-zero without changing the order

INPUT: 1 0 4 0 2 5 0 7 0 6 0

OUTPUT: 0 0 0 0 0 1 4 2 5 7 6

Your Code

```
arr = list(map(int, input().split()))

zero_count = 0
for x in arr:
    if x == 0:
        zero_count += 1

res = []
res.extend([0] * zero_count)

for x in arr:
    if x != 0:
        res.append(x)

print(*res)
```

Input

```
1 0 4 0 2 5 0 7 0 6 0
```

Output

```
0 0 0 0 0 1 4 2 5 7 6
```


Visible Test Cases

Test Case 1

Q9 **count vowels, consonants, digits, and spaces** 10 marks

Write a python program to count vowels, consonants, digits, and spaces in a given sentence

INPUT: Hello World 123

OUTPUT:

Vowels:3

Consonants:7

Digits:3

Spaces:2

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Your Code

```python
def count_chars(s):
    vowels = set("AEIOUaeiou")
    vowel_count = 0
    consonant_count = 0
    digit_count = 0
    space_count = 0

    for ch in s:
        if ch in vowels:
            vowel_count += 1
        elif ch.isalpha():
            consonant_count += 1
        elif ch.isdigit():
            digit_count += 1
        elif ch == ' ':
            space_count += 1

    return vowel_count, consonant_count, digit_count, space_count


sentence = input()
v, c, d, sp = count_chars(sentence)

print("Vowels:", v)
print("Consonants:", c)
print("Digits:", d)
print("Spaces:", sp)
```

Input

```
Hello World 123
```

Output

```
Vowels: 3
Consonants: 7
Digits: 3
Spaces: 2
```

Visible Test Cases

Test Case 1

0.00 / 10 marks

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Q10 **Palindrome** 10 marks

Write a python program to check whether the given string is a palindrome

input: malayalam

output: malayalam is a palindrome

input: saveetha

output: saveetha is not a palindrome

Your Code

```
s = input().strip()

if s == s[::-1]:
    print(f"{s} is a palindrome")
else:
    print(f"{s} is not a palindrome")
```

Input

```
saveetha
```

Output

```
saveetha is not a palindrome
```


Visible Test Cases